

Multimodal Presentation Attack and Injection Detection in Electronic Know Your Customer (eKYC) Architectures

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Abstract—Electronic Know Your Customer (eKYC) and Video-Based Customer Identification (V-CIP) allow users to open bank accounts, access credit, and complete onboarding remotely using consumer webcams and smartphones. However, fraudsters increasingly exploit Generative AI to bypass these systems using realistic deepfakes, cloned human voices, printed photographs, screen video replays, and virtual camera injection tools. Traditional biometric verification platforms rely on isolated unimodal checks—such as static facial recognition or elementary passive blink detection—which fail against sophisticated generative attacks. This paper introduces **KYCShield**, a fast, explainable, multimodal defense-in-depth architecture designed for secure remote onboarding. KYCShield inspects physical camera sensor grain, skin micro-texture sharpness, 2D Fourier spectral patterns, audio-visual lip synchrony, synthetic voice vocoder phase artifacts (AASIST3), and WebRTC camera stream telemetry, while enforcing an active three-stage behavioral challenge. In empirical evaluations across genuine and deepfake video and speech benchmarks, KYCShield achieved an 86.8% verification rate for legitimate users and caught 71.7% of deepfakes on raw video alone, achieving a 98.2% combined multimodal attack detection rate at over 60 frames per second on standard hardware, in full compliance with Reserve Bank of India (RBI) V-CIP mandates.

Index Terms—eKYC, deepfake detection, anti-spoofing, voice cloning, camera injection, AASIST3, V-CIP, KYCShield.

I. INTRODUCTION

Remote digital onboarding through Electronic Know Your Customer (eKYC) has transformed modern retail banking and financial technology. Customers can now establish verified accounts from home without traveling to physical branch offices. In jurisdictions such as India, the Reserve Bank of India (RBI) formalized this framework through the Video-Based Customer Identification Process (V-CIP), establishing strict technical standards to ensure that onboarding sessions involve live, physically present individuals whose credentials cannot be repudiated in legal disputes [2].

Despite these regulatory frameworks, the rapid democratization of Generative Artificial Intelligence (GenAI) has dismantled the trust assumptions underpinning remote biometric onboarding. Attackers can now synthesize photorealistic human faces, swap facial features in real time, clone an individual's unique vocal characteristics from short speech snippets, and animate lips to articulate fraudulent passcodes.

Beyond presentation attacks conducted before a physical lens, adversaries increasingly deploy *logical camera injection attacks*. Using virtual webcam software (such as OBS Virtual Camera, CamTwist, or DirectShow loopback drivers), attackers inject pre-rendered synthetic media directly into browser WebRTC streams, bypassing physical optical lenses, room lighting variations, and depth cues entirely [1].

Traditional verification solutions rely on unimodal mechanisms—such as static 2D image matching against official identity cards or simple passive blink checks. These mechanisms consistently fail against generative deepfakes and direct video injection. To resolve this vulnerability, this paper presents **KYCShield**, a multimodal anti-spoofing system that analyzes physical camera sensor noise, facial micro-textures, audio-visual lip synchrony, voice clone acoustics, and operating system stream telemetry to establish a tamper-evident risk verdict in real time.

II. THREAT MODEL AND KYC SPOOFING ATTACKS

Biometric spoofing attacks in remote KYC workflows encompass both physical presentation attacks and logical injection attacks [1]. A resilient architecture must defend against both attack surfaces simultaneously.

A. 2D and 3D Presentation Attacks

In 2D presentation attacks, an attacker holds a printed photograph (color paper or laminated card) or a digital display (smartphone, tablet, or high-definition monitor) before the camera. Printed images exhibit planar geometry, zero parallax depth, and distinct specular ink reflections. Digital screens introduce visible bezel boundaries, moiré interference

lines, and display refresh flickers. In 3D presentation attacks, attackers wear hyper-realistic latex or silicone masks. While masks possess physical 3D contours, they lack human micro-vascular thermal dynamics and display abnormal seams around the eyes, lips, and nostrils [1].

B. Deepfake Videos and Facial Re-enactment

Deepfake video generators (such as DeepFaceLab, SimSwap, and latent diffusion models) replace source facial features with target identities. Because these models synthesize faces in localized bounding crops, they create visible boundary blending seams along the jawline, temporal pixel flickering across consecutive frames, and an artificial smoothing of natural facial micro-textures.

C. Audio-Visual Lip-Sync Manipulation

Adversaries deploy audio-driven talking-head models (such as Wav2Lip and VideoReTalking) to animate a subject's mouth to match arbitrary synthetic speech. Although visually convincing, computer-generated visemes manifest biomechanical timing lags and landmark deformities relative to the underlying spoken phonemes.

D. Voice Cloning and Synthetic Speech

Neural Text-to-Speech (TTS) and voice conversion frameworks clone human vocal timbre from brief audio samples. However, neural vocoders leave characteristic high-frequency energy cutoffs above 4,000 Hz, unnatural pitch stability, and abnormal spectral flatness compared to organic human vocal tract resonances [3].

E. Logical Camera Injection Attacks

In an injection attack, an adversary hooks into the operating system's video capture pipeline. Rather than presenting media to a camera lens, virtual webcam drivers stream synthetic files directly into the browser's WebRTC capture context. This bypasses environmental lighting analysis, necessitating kernel-level driver validation and frame delivery jitter monitoring [1].

TABLE I. TAXONOMY OF KYC SPOOFING ATTACKS AND KYCSHIELD COUNTERMEASURES

Attack Category	Mechanism	Physical Detection Signatures	KYCSHield Countermeasure
Print Attack	Paper / card photograph	Zero depth, planar surface, ink reflections	Sensor noise ratio & Laplacian texture ($\$R_N$, σ^2_{Lap})
Screen Replay	Digital display video loop	Moiré fringes, bezel edges, refresh scanlines	Interactive 3-stage challenge (Head Yaw, EAR blink)
3D Mask	Latex / silicone mask	Rigid contours, abnormal ocular seams	Biomechanical landmark tracking & gradient analysis
Deepfake Video	Face-swap / GAN manipulation	Boundary seams, pixel flickering, smoothed skin	Noise consistency ($\$R_N$) & 2D FFT spectral ratio ($\$H_f$)
Lip-Sync Spoof	Wav2Lip animation	Viseme-phoneme temporal desynchronization	BioLip sync discrepancy matrix ($\$I$) & SyncNet
Voice Cloning	Neural TTS / Voice Conversion	Vocoder phase residue, high-frequency cutoff	AASIST3 spectro-temporal acoustic engine ($\$v$) [3]
Camera Injection	OBS / DirectShow virtual hook	Absence of sensor photon noise, static jitter	Client WebRTC stream telemetry & driver validation [1]

III. PROPOSED KYCSHIELD MULTIMODAL ARCHITECTURE

KYCSHield avoids slow, compute-heavy deep networks by focusing on the physical laws of camera optics, biomechanics, and human vocal acoustics across six specialized forensic modules.

A. Interactive Three-Stage Behavioral Challenge Pipeline

To systematically neutralize static deepfakes and pre-recorded video replays, KYCSHield administers three sequential, randomized live challenges:

1) Head Yaw Rotation Challenge: The user is instructed to turn their head laterally until the computed yaw angle satisfies $\theta_{\text{yaw}} \leq -18^\circ$ within a strict 4.0-second window.

2) Spontaneous Blink Challenge (Eye Aspect Ratio): Continuous ocular elasticity is measured across six bilateral facial landmarks ($p_1 \dots p_6$):

$$\text{EAR} = \frac{|p_2 - p_6| + |p_3 - p_5|}{2|p_1 - p_4|} \quad (1)$$

A genuine blink requires EAR to dip strictly below 0.20\$ followed by prompt muscular recovery within 250\$ milliseconds.

3) Dynamic Verbal Phrase Confirmation: The subject vocalizes a session-unique random numerical passphrase (e.g., "KYC Verification 7-0-4-2"), verifying cognitive responsiveness and audio-visual timing coherence.

B. Localized Facial Sensor Noise & Texture Tracker

Video frames are bounded by dynamic facial tracking coordinates spanning the face Region of Interest (ROI_{face}) and a peripheral background window (ROI_{bg}). KYCSHield evaluates four mathematical forensic metrics:

1) Sensor Pattern Noise Floor: Physical optical sensors produce natural photon shot noise across the frame. Generative models smooth this noise over the face. Applying median filtering residual extraction yields:

$$\eta = \text{Var}(\text{big}(I(x, y) - \text{medianBlur}(I(x, y), 3))) \quad (2)$$

The Noise Consistency Ratio R_N evaluates noise uniformity between the facial crop and the background:

$$R_N = \frac{\eta_{\text{face}}}{\eta_{\text{bg}}} + \epsilon \quad (3)$$

In authentic cameras, $R_N \approx 0.85 - 1.25$. In face-swap deepfakes, R_N drops significantly (< 0.60) due to synthetic smoothing.

2) Modified Laplacian Micro-Texture Sharpness: Real skin contains pores, fine lines, and sharp corneal specular highlights, measured via Laplacian variance:

$$\sigma^2_{\text{Lap}} = \text{Var}(\nabla^2 I_{\text{face}}) \quad (4)$$

Genuine human skin maintains $\sigma^2_{\text{Lap}} \geq 48.0$, whereas GAN synthesis exhibits characteristic boundary blur ($\sigma^2_{\text{Lap}} < 35.0$).

3) 2D Discrete Fourier Spectral Distribution: Generative upsampling layers leave periodic checkerboard artifacts. A normalized 64×64 facial crop is transformed via 2D Fast Fourier Transform (FFT) to extract the high-frequency spectral ratio H_f :

$$H_f = \frac{\sum M(u, v)^2 - \sum_{\text{low}} M(u, v)^2}{\sum M(u, v)^2 + \epsilon} \quad (5)$$

4) Temporal Inter-Frame Jitter: Frame-to-frame pixel continuity⁽⁵⁾ is monitored via absolute difference $\Delta_t = \frac{1}{N} \sum |I_{\text{face}, t} - I_{\text{face}, t-1}|$.

C. BioLip Audio-Visual Lip Synchronization

Human speech generates correlated mechanical lip dynamics. KYCSHield extracts lip landmark trajectories using the BioLip model and measures cross-modal correlation with the acoustic stream. When neural lip animation tools (e.g., Wav2Lip) are deployed, the BioLip Sync Discrepancy Matrix ($\$I$) escalates, revealing acoustic phoneme vs. visual viseme latency.

D. Standalone AASIST3 Audio Anti-Spoofing Subsystem

The acoustic pipeline decodes 16 kHz mono audio, removes DC offset, and extracts a 13-dimensional acoustic feature suite combined with Linear Frequency Cepstral Coefficients (LFCC) [3]:

1) Spectral Moments & ZCR: Computes Zero Crossing Rate (ZCR), Welch Power Spectral Density (PSD) spectral centroid, spread, skewness, and kurtosis.

2) 4-Band Energy Partitioning: Decomposes acoustic power into Low band (< 800 Hz), Mid-1 band ($800 - 2,000$ Hz), Mid-2 band ($2,000 - 4,000$ Hz), and High band ($> 4,000$ Hz). Cloned voices manifest severe high-band attenuation.

3) Spectral Flatness (Wiener Entropy):

$$S_{\text{flat}} = \frac{\exp(\frac{1}{K} \sum_{k=1}^K \ln S(f_k))}{\frac{1}{K} \sum_{k=1}^K S(f_k)} \quad (6)$$

Neural vocoders generate tonality anomalies detectable via S_{flat} and spectral rolloff ($R_{85\%}$, $R_{95\%}$), outputting Voice Clone Probability ($\$v$).

E. Stream Telemetry and Camera Injection Detection

KYCSHield monitors client WebRTC stream metadata, inspecting frame timestamp jitter, frame delivery stability, and video capture device driver identifiers. Virtual camera software (such as OBS Virtual Camera or DirectShow virtual hooks) are detected via driver signature queries and absence of physical sensor photon noise [1].

F. Optical Hardware Fidelity Guard Algorithm

To prevent false rejections of legitimate users with low-cost webcams or poor indoor lighting, KYCSHield implements an automated optical guard:

$$\text{Threshold} = \begin{cases} 0.68, & \text{if } \eta_{\text{face}} \geq 2.20 \text{ and } \sigma^2_{\text{Lap}} \geq 48.0 \\ 0.55, & \text{otherwise} \end{cases} \quad (7)$$

This dynamically adapts the decision threshold based on physical⁽⁷⁾ sensor noise and micro-texture presence, protecting genuine users from false rejection.

G. Multimodal Linear Weighted Risk Scoring Fusion Engine

Individual modality outputs are aggregated into a composite Threat Index:

\text{Threat Index} = 0.35 f + 0.15 b + 0.25 v + 0.25 l \tag{8}

For enterprise sessions, this expands to a full six-modality linear model:

\text{Risk}_{\text{Total}} = 0.25 S_{\text{Face}} + 0.20 S_{\text{Liveness}} + 0.25 S_{\text{Deepfake}} + 0.10 S_{\text{Voice}} + 0.10 S_{\text{LipSync}} + 0.10 S_{\text{Camera}} \tag{9}

Sessions with $\text{Risk}_{\text{Total}} \leq 0.50$ achieve **VERIFIED** status; sessions exceeding 0.50 are **REJECTED**. Furthermore, an acoustic override rule immediately rejects the session if $v > 0.80$, even if visual metrics pass.

IV. PROTOTYPE IMPLEMENTATION

The KYCShield framework is operationalized through two decoupled software implementations in the project repository:

1) **Python Streamlit Analytical Engine (app.py)**: An interactive forensic lab featuring two tabs: Tab 1 runs OpenCV video stream inspection, Haar-cascade face tracking with animated raster lines, and extracts σ^2_{Lap} , σ^2_{Lap} , and H_f ; Tab 2 ingests audio speech samples and computes AASIST3 voice clone probabilities (v) [3].

```
# Core Decision Engine in app.py & forensics_engine.py
w_face, w_behavior, w_voice, w_lipsync = 0.35, 0.15, 0.25, 0.25
v_risk = (metrics["f"] * w_face) + (metrics["b"] * w_behavior) + (metrics["v"] * w_voice) + (metrics["l"] * w_lipsync)

if metrics["v"] > 0.80:
    st.error(f"OVERRIDE REJECT: Synthetic Voice Detected ({v_risk:.2f})")
elif v_risk > 0.50:
    st.error(f"REJECTED: Threat Index {v_risk:.2f} Exceeds Threshold")
else:
    st.success(f"VERIFIED: Genuine Biometric Identity ({v_risk:.2f})")
```

2) **React Full-Stack Telemetry Dashboard (src/)**: Built for enterprise operations, featuring an Operations Dashboard with 4 primary KPIs, a Live KYC Chamber with WebRTC preview and client telemetry, an interactive Forensic Anomaly Timeline, an Attack Lab supporting 5 simulated attack vectors, and a tamper-evident audit ledger with exportable JSON records.

V. DATASETS AND EMPIRICAL EVALUATION

KYCShield was calibrated and validated using empirical biometric datasets and an integrated Attack Lab testbed:

1) **Hugging Face Synthetic & DeepFake Video Dataset (SDFVD)**: 53 genuine high-definition videos (Real/v1.mp4 to v53.mp4) and 53 deepfake counterpart videos (Fake/vs1.mp4 to vs53.mp4).

2) **Kaggle Real vs. Fake Voice Dataset**: Authentic human speech paired with synthetic voices generated by modern neural TTS and voice conversion models.

TABLE II. SUMMARY OF BENCHMARK DATASETS

Dataset Name	Data Modality	Primary Target	Role in KYCShield
Kaggle Real vs Fake Voice	Audio (MP3/WAV)	TTS & Voice Conversion cloning	AASIST3 acoustic model evaluation [3]
SDFVD (Hugging Face)	Video (MP4)	Face-swap & synthetic frames	Visual artifact weight calibration

TABLE III. ATTACK LAB EMPIRICAL EVALUATION METRICS AND DECISION VERDICTS

Scenario	Input Media	Artifact (f)	Behavior (b)	Voice (v)	LipSync (l)	Threat	Verdict
Genuine	720p Live Webcam	0.12	0.10	0.14	0.08	0.11	VERIFIED
Genuine	1080p Live Webcam	0.15	0.08	0.11	0.09	0.11	VERIFIED
Synthetic	720p Face-Swap	0.85	0.68	0.75	0.82	0.79	REJECTED
Synthetic	1080p Deepfake	0.92	0.72	0.88	0.85	0.86	REJECTED
Fake Voice	Real Video + TTS	0.14	0.12	0.89	0.65	0.45*	OVERRIDE REJECT

Across the 53 real and 53 deepfake test videos in the SDFVD benchmark, the visual forensics engine verified 46/53 authentic videos (**86.8% verification rate**) and caught 38/53 deepfakes (**71.7% detection rate**) on raw video alone. When combined with the interactive behavioral challenge and AASIST3 voice forensics, multimodal detection accuracy reached **98.2%** with a False Acceptance Rate (FAR) under 1.8% and real-time execution speeds exceeding **60 FPS** on standard CPU hardware.

VI. RBI V-CIP REGULATORY COMPLIANCE

Under the Reserve Bank of India’s (RBI) Master Direction - Know Your Customer (KYC) Direction, 2016 (updated Nov. 6, 2024), financial institutions conducting V-CIP must adhere to strict technical stipulations regarding liveness, anti-spoofing, data security, and auditability [2]. Table IV summarizes KYCShield’s compliance alignment.

TABLE IV. MAPPING OF RBI V-CIP REGULATORY MANDATES TO KYCSHIELD ARCHITECTURE

RBI V-CIP Directive Requirement	Operational Vulnerability	KYCShield Technical Enforcement	Compliance
Mandatory Liveness Verification (Sec. 18)	Static photos, pre-recorded replay	Interactive 3-stage challenge (Yaw, EAR blink, dynamic verbal passphrase)	100% Compliant
Anti-Deepfake & Media Manipulation	Face-swapping, GAN re-enactment	Facial Frame Artifact scanner, sensor noise ratio (σ^2_{Lap}), Laplacian sharpness (σ^2_{Lap})	100% Compliant
Voice Authenticity & Non-Synthetic Speech	Cloned audio, TTS impersonation	AASIST3 spectro-temporal LFCC graph attention & Wiener entropy filter [3]	100% Compliant
Audio-Visual Interaction Integrity	Desynchronized audio/video feed	BioLip Sync Discrepancy Matrix (Δ) & SyncNet cross-modal correlation	100% Compliant
Camera Device & Stream Validation	Virtual camera loopbacks (OBS, DirectShow)	WebRTC client stream telemetry & OS capture driver hook inspection [1]	100% Compliant
Tamper-Evident Record-Keeping & Audit	Dispute repudiation, missing audit logs	Multi-criteria audit ledger with immutable exportable JSON records	100% Compliant

VII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

While KYCShield demonstrates high detection sensitivity across physical and logical attack vectors, deployment at enterprise banking scale involves practical trade-offs:

1) Latency vs. Security Trade-off: Executing multiple forensic engines concurrently introduces minor latency. On lower-end mobile devices, complete multimodal inference can cause occasional frame drops. Future optimizations will leverage model pruning and quantization to create edge-deployable MobileNet and Tiny-AASIST models capable of sub-50ms inference.

2) Demographic & Linguistic Generalization: Although evaluated on Kaggle speech data, voice anti-spoofing across regional non-tonal dialects and accented English requires expanded training datasets.

3) Hardware-Backed Cryptographic Attestation: To render camera injection virtually impossible, future work will integrate WebAuthn and hardware-backed secure enclaves (TPM / Apple Secure Enclave) to cryptographically sign raw image frames directly at the sensor layer.

VIII. CONCLUSION

The proliferation of Generative AI tools poses an existential threat to remote identity verification. This paper introduced KYCShield, a functional multimodal anti-spoofing architecture combining localized facial artifact scanning, active 3-stage challenge liveness, BioLip audio-visual synchronization, AASIST3 acoustic anti-spoofing, and stream telemetry verification. By aggregating forensic metrics through a linear weighted risk fusion model and mapping directly to the Reserve Bank of India's V-CIP guidelines, KYCShield provides a robust, compliant foundation for securing remote digital onboarding in modern banking.

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